

PRACTICAL COURSE WORKBOOK

Python for AI & Data Science

Turn untidy data into a defensible finding

LEVEL	Beginner
GUIDED TIME	5 hours across 12 lessons
PROJECT	a cleaned dataset, analysis and decision-ready chart
SKILLS	Python, NumPy, Pandas, Matplotlib, Critical evaluation, Documentation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use Python appropriately in a realistic, bounded task.
- Use NumPy appropriately in a realistic, bounded task.
- Use Pandas appropriately in a realistic, bounded task.
- Use Matplotlib appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners using data and code to answer practical questions.

BEFORE YOU BEGIN

- Basic numeracy and file-management skills

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Python for AI & Data Science is a structured, practical course covering Python, NumPy, Pandas, Matplotlib. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Python: Data foundations	1. Problem framing with Python 2. Data types with NumPy 3. Tool setup with Pandas
02 NumPy: Analysis	1. Cleaning with Matplotlib 2. Exploration with Python 3. Modelling with NumPy
03 Pandas: Evaluation	1. Validation with Pandas 2. Uncertainty with Matplotlib 3. Communication with Python
04 Matplotlib: Applied project	1. Workflow with NumPy 2. Review with Pandas 3. Reproducibility with Matplotlib

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Python: Data foundations

Apply Python through problem framing, data types, tool setup.

LESSON 01 / 18 MIN

Problem framing with Python

Learning objectives

- Explain problem framing in the context of Python for AI & Data Science.
- Apply Python to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

problem framing, Python, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a problem framing result produced with Python?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Data types with NumPy

Learning objectives

- Explain data types in the context of Python for AI & Data Science.
- Apply NumPy to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

data types, NumPy, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a data types result produced with NumPy?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Tool setup with Pandas

Learning objectives

- Explain tool setup in the context of Python for AI & Data Science.
- Apply Pandas to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

tool setup, Pandas, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a tool setup result produced with Pandas?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

NumPy: Analysis

Apply NumPy through cleaning, exploration, modelling.

LESSON 04 / 30 MIN

Cleaning with Matplotlib

Learning objectives

- Explain cleaning in the context of Python for AI & Data Science.
- Apply Matplotlib to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

cleaning, Matplotlib, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a cleaning result produced with Matplotlib?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Exploration with Python

Learning objectives

- Explain exploration in the context of Python for AI & Data Science.
- Apply Python to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

exploration, Python, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a exploration result produced with Python?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Modelling with NumPy

Learning objectives

- Explain modelling in the context of Python for AI & Data Science.
- Apply NumPy to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

modelling, NumPy, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a modelling result produced with NumPy?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Pandas: Evaluation

Apply Pandas through validation, uncertainty, communication.

LESSON 07 / 26 MIN

Validation with Pandas

Learning objectives

- Explain validation in the context of Python for AI & Data Science.
- Apply Pandas to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

validation, Pandas, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a validation result produced with Pandas?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Uncertainty with Matplotlib

Learning objectives

- Explain uncertainty in the context of Python for AI & Data Science.
- Apply Matplotlib to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

uncertainty, Matplotlib, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a uncertainty result produced with Matplotlib?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Communication with Python

Learning objectives

- Explain communication in the context of Python for AI & Data Science.
- Apply Python to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

communication, Python, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a communication result produced with Python?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Matplotlib: Applied project

Apply Matplotlib through workflow, review, reproducibility.

LESSON 10 / 22 MIN

Workflow with NumPy

Learning objectives

- Explain workflow in the context of Python for AI & Data Science.
- Apply NumPy to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

workflow, NumPy, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a workflow result produced with NumPy?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 11 / 26 MIN

Review with Pandas

Learning objectives

- Explain review in the context of Python for AI & Data Science.
- Apply Pandas to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

review, Pandas, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a review result produced with Pandas?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 12 / 30 MIN

Reproducibility with Matplotlib

Learning objectives

- Explain reproducibility in the context of Python for AI & Data Science.
- Apply Matplotlib to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

reproducibility, Matplotlib, data

Retrieval prompt

In Python for AI & Data Science, which evidence best supports a reproducibility result produced with Matplotlib?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Python for AI & Data Science project using Python, NumPy, Pandas.

DELIVERABLE	A working Python for AI & Data Science artefact plus an evidence-based self-review.
TOOLS	A suitable Python environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Python.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

The Python Tutorial

Python Software Foundation - reviewed 2026-08-21

<https://docs.python.org/3/tutorial/>

User Guide

scikit-learn - reviewed 2026-08-21

https://scikit-learn.org/stable/user_guide.html

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Python scope.